

Thomas Hong

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EDUCATION

The Ohio State University

Bachelor of Science in Electrical and Computer Engineering

Columbus, OH

Aug. 2025 – Dec 2028

EXPERIENCE

Electrical & Instrumentation Engineering Co-op

Marathon Petroleum Corporation

Aug. 2026 – Present

Detroit, MI

- Support instrumentation engineering for a refinery processing 146,000 barrels/day, implementing wired and wireless monitoring solutions for pressure, temperature, valve position, and other process parameters
- Modify AutoCAD electrical and instrumentation schematics for 13 planned projects spanning wireless, wired, and existing system revisions, documenting device locations, wiring, and installation requirements
- Work 50/50 between engineering office and refinery field, coordinating with Field Services on device placement and providing installation instructions to support instrumentation deployment

Engineering Intern – Transmission Strategy and Grid

American Electric Power

May 2026 – Aug 2026

New Albany, OH

- Conduct transmission infrastructure vulnerability assessments on transformers, lines, and circuits within a 70,000+ asset database, evaluating each against operational standards across AEP's 11-state grid
- Identified and led resolution of two critical SQL defects in the Copperleaf system: negative OMS loads tied to Indiana and Texas solar stations, and broken customer count, CI, and CMI metrics caused by source code errors
- Contribute to an automated Python pipeline to migrate legacy asset data into the new system, replacing a manual process currently required for 6 of 10 device types and reducing transcription errors

Undergraduate Research Assistant – MITEC Process Team

Nanotech West Laboratory – The Ohio State University

Oct. 2025 – Present

Columbus, OH

- Process 20+ silicon wafers per run through full cleanroom photolithography cycles using spin coating, mask alignment, and controlled bake processes, sustaining 0 contamination incidents across 5+ months and ensuring uninterrupted fabrication
- Detect pattern fidelity defects across 20+ runs using stylus profilometry and optical microscopy, reducing reprocessing cycles by identifying process drift pre-etch and improving process consistency
- Standardize process documentation and optimize resist thickness, bake timing, and mask alignment, cutting researcher onboarding time by 30% and maintaining consistent critical dimensions across runs

Boeing Engineering Project Internship – IgnisKey

The Boeing Company

Feb. 2025 – Apr. 2025

Ridley Park, PA

- Led a 4-member engineering team through 5 prototype iterations, delivering a functional hardware prototype and formal design review to Boeing engineers on schedule and within scope
- Reduced PCB build complexity across 5 prototype cycles, shortening assembly time each iteration and eliminating redundant steps identified in failure-mode reviews
- Integrated competing thermal and ergonomic constraints into a unified system-level design, producing trade-off analyses and prototype documentation cited by Boeing engineers as a model

Freelance PC Builder and Consultant

Hong Computing

Mar. 2021 – Present

Upper Darby, PA

- Design and deliver 200+ custom desktop systems, managing lifecycle from sourcing to assembly and validation
- Diagnose and resolve complex hardware faults, including power delivery instability, memory timing failures, and thermal throttling, reducing mean time-to-resolution through systematic root-cause analysis
- Engineer custom air and liquid cooling solutions that reduce peak CPU and GPU temperatures by 15–25°C in thermally constrained cases, extending component lifespan and improving sustained performance

PROJECTS

Dynamic Transmission Line Monitoring System – 2nd Place

AEP Dynamic Grid Challenge – MakeOHI/O 2026

Mar. 2026

Columbus, OH

- Designed and built a real-time transmission line monitoring prototype in 24 hours using an Arduino Nano ESP32, VL53L4CD LiDAR for sag measurement, and DS18B20 temperature sensor
- Analyzed 700+ sensor measurements exported as CSV data, producing visualizations that quantified the conductor temperature-to-sag relationship and matched IEEE thermal expansion models within a 5% error margin
- Simulated forced-wind cooling to study convection effects on conductor temperature, a variable most competing teams did not model

TECHNICAL SKILLS

Languages: Java, Python, C/C++, JavaScript, MATLAB, SQL

Hardware: PCB design, schematic capture, embedded systems, microcontrollers, FPGA, OrCAD, Cadence, hardware validation

Lab Skills: Cleanroom operations, photolithography, wafer preparation, sample handling documentation

Software & Tools: Linux, Windows 10/11, Microsoft 365, Visual Studio, Excel, SharePoint, Arduino IDE, Copperleaf

CERTIFICATIONS

Learning PCB Design with OrCAD | LinkedIn Learning

Apr. 2026

Computer Architecture Essentials | LinkedIn Learning

Apr. 2026